Pod

Lead, camera header, 3.3 V, light

File: pod.kicad_sch

Camera pod v1.0 — PH 8 lead from the bay node's J6, 1 × 6 header for the Arducam Mega's cable,
  AP2114H 3.3 V for the camera, four white 2835 LEDs switched by the node (LED_K).
Lead pinout here: 1 GND · 2 CS · 3 LED_K · 4 MISO · 5 MOSI · 6 CAM_5V · 7 SCK · 8 GND
  (the node's J6 mirrored: one 'Same direction' PH 8 cable).

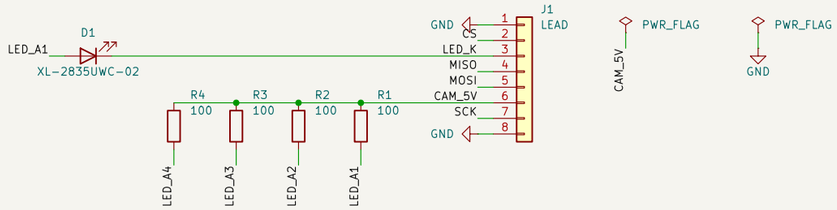
Generated from electronics/camera-pod/design.py — edit the Python, then regenerate
The bay camera pod's board: lead in, the Arducam Mega's header, its 3.3 V, the light (hw-testing-rac
hw-testing-rack

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File: camera-pod.kicad_sch

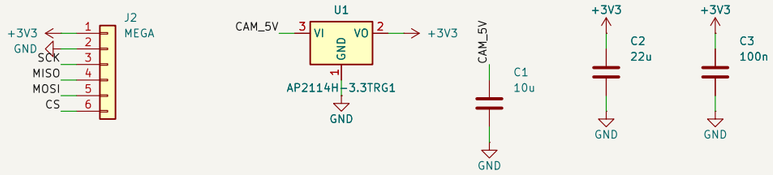
Title: Camera pod — Overview

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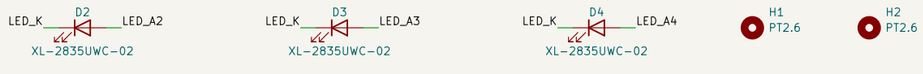
Lead from the bay node's J6 (JST PH 8, side entry)



The Arducam Mega's cable (1 × 6, 2.54 mm, its own order) and its 3.3 V



Light: four white LEDs, 100 Ω each from CAM_5V, cathodes to LED_K (the node's low side)



Test pads (bottom side, labelled): no short to GND, then CAM_5V and 3.3 V at power-up



J1 is the bay node's J6 mirrored: a 'Same direction' PH 8 lead (pin n to pin 9−n) joins them.
J2 follows the Mega's own pin order; the names are on the silkscreen, its cable plugs on by name.
The Mega from 3.3 V (U1, ≈ 0.5 W at its 317 mA max, Arducam): its MISO is 3.3 V. LEDs 18-26 mA each (100 Ω 0805, ≤ 66 mW),
  ≤ 102 mA for the four, off unless the node's AO3400A pulls LED_K low (PWM ≥ 20 kHz or on/off).
CAM_5V: the node's 0.75 A PTC, no switch: a hung camera (the Mega has no reset pin) needs a power cycle.

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hw-testing-rack

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Title: Camera pod — Lead, camera header, 3.3 V, light		
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